■ Developer Mock Test Results

Student: **sameena shaik**

Name:	sameena shaik	**Student ID:**	118
Date:	2026-03-05 12:38:00	**Performance:**	■■ Needs Improvement
Overall Score:	9.0/25 (36.0%)	**Test ID:**	226cd751-6e45-43e9...
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	2/10	20.0%	■■ Needs Work
■ MCQ/Theory	2/10	20.0%	■■ Needs Work
■ Coding	5.0/5	100.0%	■ Pass

■ Detailed Question Review

■ APTITUDE SECTION (2/10 - 20.0%)

■ Q1. A mixture is made of 2 parts oil and 5 parts water. What is the ratio of oil to the total mixture?

Your Answer: 2/7

■ *Excellent! Right answer.*

■ Q2. What is the next number: 3, 6, 9, 12, 15, ?

Your Answer: 22

Correct Answer: 18

■ *The correct answer is 18 because the pattern in the sequence is adding 3 to the previous number. The key calculation step is: 15 (previous number) + 3 = 18. The user's likely mistake was not recognizing the consistent addition of 3 in the sequence, instead possibly looking for a different pattern or adding a different number.*

■ Q3. If a number is decreased by 20% and then increased by 20%, what is the overall percentage change?

Your Answer: 10%

Correct Answer: -4%

■ *The correct answer is -4% because when a number is decreased by 20% and then increased by 20%, the overall effect is a decrease. The key calculation step is: if we start with 100, decreasing by 20% gives 80, and then increasing 80 by 20% gives 96, which is 4% less than the original 100. The user likely made a mistake by not considering the effect of the percentage changes on the reduced value, rather than the original value.*

■ Q4. If 5 men can build a wall in 10 days, how many men are needed to build the same wall in 5 days?

Your Answer: 6

Correct Answer: 10

■ *To determine the number of men needed to build the wall in 5 days, we can use the formula: men * days = constant. Given that 5 men can build the wall in 10 days, we have 5 * 10 = 50 man-days. To build the same wall in 5 days, we need 50 man-days / 5 days = 10 men. The user's answer of 6 men is likely incorrect because it doesn't account for the proportional increase in manpower needed to halve the construction time.*

■ Q5. What is the next number: 1, 4, 9, 16, 25, ?

Your Answer: 42

Correct Answer: 36

■ *The correct answer is 36 because the pattern of the sequence is obtained by squaring consecutive integers: 1^2 = 1, 2^2 = 4, 3^2 = 9, 4^2 = 16, 5^2 = 25, and 6^2 = 36. The user's answer, 42, is likely incorrect because it doesn't follow this pattern. The key calculation step is squaring the next consecutive integer, which is 6^2 = 36.*

■ Q6. If it is true that all books are written and all articles are written, what can be concluded about books and articles?

Your Answer: They are both written

■ *Correct! Well done.*

■ Q7. If 15 workers can build a house in 10 days, how many days will it take for 12 workers to build the same house?

Your Answer: 14 days

Correct Answer: 12 days

■ *To find the number of days it will take for 12 workers to build the house, we can use the formula: (number of workers initial) * (number of days initial) = (number of workers final) * (number of days final). So, 15 workers * 10 days = 12 workers * x days. The key calculation step is: 15 * 10 = 12 * x, which simplifies to 150 = 12x, and then x = 150 / 12 = 12.5. The user's likely mistake is rounding or incorrectly calculating the result of this division. The correct answer should be around 12.5 days, which is closer to 12 days than 14 days, given the provided options.*

■ Q8. A store buys a product for $40 and sells it for $50. What is the profit percentage?

Your Answer: 20%

Correct Answer: 25%

■ *To find the profit percentage, we need to calculate the profit made and then divide it by the cost price. The key calculation step is: Profit = Selling Price - Cost Price = $50 - $40 = $10, and then Profit Percentage = (Profit / Cost Price) * 100 = ($10 / $40) * 100 = 25%. The user's answer of 20% is likely incorrect because they may have made a calculation error, such as incorrectly calculating the profit or percentage.*

■ Q9. If it is true that all A are B and some B are C, what can be concluded about A and C?

Your Answer: No A are C

Correct Answer: Some A are C

■ *The correct answer is "Some A are C" because if all A are B and some B are C, then it's possible that the B's that are also C overlap with the A's that are B, resulting in some A being C. The user's answer, "No A are C", is incorrect because it assumes that the groups A and C are completely separate, which isn't necessarily true based on the given information. The user likely made a mistake by not considering the potential overlap between the groups A, B, and C.*

■ Q10. If 6 workers can build a house in 10 days, how many workers will be needed to build the same house in 8 days?

Your Answer: 9 workers

Correct Answer: 8 workers

■ *To find the number of workers needed to build the house in 8 days, we can use the formula: workers * days = constant. Given that 6 workers can build the house in 10 days, we have 6 * 10 = 60 worker-days. To build the same house in 8 days, we need 60 worker-days / 8 days = 7.5 workers. The user's answer of 9 workers is likely a mistake due to overestimation. The correct calculation shows that approximately 7.5 workers are needed, which is closer to option A) 7 workers or B) 8 workers, but since we can't have a fraction of a worker and the question likely seeks a whole number, the closest whole number to 7.5 is indeed 8 workers, which matches option B.*

■ MCQ SECTION (2/10 - 20.0%)

■ Q1. What is a method in Java?

Your Answer: A reusable block of code that performs a task

■ *Correct! Good understanding.*

■ Q2. What are the types of data that can be stored in a variable in Java?

Your Answer: Int, double, char, String, arrays

■ *Excellent! Right answer.*

■ Q3. What is multi-threading in Java?

Your Answer: The ability of a program to execute a single thread

Correct Answer: The ability of a program to execute multiple threads concurrently

■ *The correct answer is A) The ability of a program to execute multiple threads concurrently, because multi-threading in Java allows a program to run multiple threads simultaneously, improving responsiveness and system utilization. The user's answer, B) The ability of a program to execute a single thread, is incorrect because it describes a single-threaded program, not a multi-threaded one. The user's likely mistake is confusing the concept of multi-threading with single-threading, which are two distinct programming paradigms in Java.*

■ Q4. What is an example of a simple Java program?

Your Answer: A program that prints a greeting message

Correct Answer: All of the above

■ *The correct answer is "All of the above" because a simple Java program can indeed be any of these options, as they all involve basic programming concepts. The user's answer, "A program that prints a greeting message", is a correct example of a simple Java program, but it is not the most comprehensive answer. The user's likely mistake was underestimating the breadth of possibilities for a simple Java program, which can include a wide range of basic applications such as calculating student grades, managing simple data, or printing a greeting message.*

■ Q5. What is a characteristic of Java?

Your Answer: Object-based

Correct Answer: Platform-independent

■ *The correct answer is B) Platform-independent because Java is designed to be compiled into an intermediate format called bytecode, which can be run on any platform that has a Java Virtual Machine (JVM) installed. This allows Java programs to be executed on multiple operating systems without the need for recompilation. The user's answer, "Object-based", is likely incorrect because while Java is an object-oriented language, the question is asking about a specific characteristic of Java, and "object-based" is not as accurate a description of Java's platform independence.*

■ Q6. How can a Java program be executed?

Your Answer: Using a web browser

Correct Answer: Using an IDE

■ *The correct answer is actually not just "Using an IDE", but also includes "Using the command prompt" as Java programs can be executed from the command line using the `java` command. The user's answer, "Using a web browser", is incorrect because Java programs are typically compiled and run on the local machine, not executed directly in a web browser. The user's likely mistake is confusing Java with JavaScript, which can be executed in a web browser, or not understanding the typical execution environment for Java programs.*

■ Q7. What is one of the core concepts in Java used to store data values?

Your Answer: Classes

Correct Answer: Variables

■ *The correct answer is Variables because they are the core concept in Java used to store data values. Variables are essentially named storage locations that hold values, allowing programmers to use and manipulate the data in their code. The user's answer, Classes, is likely incorrect because while classes are a fundamental concept in Java, they are used to define blueprints for objects, not to store individual data values.*

■ Q8. What do methods help organize code into?

Your Answer: Classes
Correct Answer: Reusable parts
■ *The correct answer is "Reusable parts" because methods help organize code into reusable blocks that can be called multiple times from different parts of a program, reducing code duplication and improving maintainability. The user's answer, "Classes", is incorrect because while methods are often defined inside classes, they themselves help organize code into reusable parts, not classes. The user likely mistook the container (class) for the purpose of the method (reusability).*

■ Q9. What is exception handling in Java?

Your Answer: The process of handling logical errors
Correct Answer: The process of handling runtime errors
■ *The correct answer is A) The process of handling runtime errors, because exception handling in Java is a mechanism that allows developers to manage and respond to runtime errors, also known as exceptions, which occur during the execution of a program. The user's answer, C) The process of handling logical errors, is incorrect because logical errors are typically caught during the testing and debugging phase, and are not directly related to exception handling. The user likely mistook exception handling with debugging or testing, which are separate processes that help identify and fix logical errors.*

■ Q10. What is required to run Java programs?

Your Answer: Java SDK
Correct Answer: Java JDK
■ *The correct answer is Java JDK (Java Development Kit) because it includes the Java Runtime Environment (JRE) and development tools, which are necessary to compile and run Java programs. The user's answer, Java SDK, is likely incorrect because SDK (Software Development Kit) is an older term that was used for Java versions prior to Java 10, and it is now referred to as JDK. The JDK provides the necessary tools to develop, compile, and run Java programs, making it the required component.*

■ CODING SECTION (5.0/5 - 100.0%)

■ Q1. Write a Java program that reads a stack of integers from stdin and finds the maximum element. **Input Format:** - Line 1: The number of elements in t...

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();   // number of elements
        Stack<Integer> stack = new Stack<>();

        for(int i = 0; i < n; i++) {
            stack.push(sc.nextInt());
        }

        int max = Integer.MIN_VALUE;

        while(!stack.isEmpty()) {
            int value = stack.pop();
            if(value > max) {
                max = value;
            }
        }
```

```
        System.out.println(max);
    }
}
```

■ *Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving an "Accepted" status, indicating that your code is correct and effective.*

■ Q2. Write a program that reads two integers and prints their greatest common divisor (GCD). **Input Format:** - Line 1: First integer (integer) - Line 2:...

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();

        // Euclidean Algorithm
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }

        System.out.println(a);
    }
}
```

■ *Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and a well-implemented solution. Great job on achieving an "Accepted" status, indicating that your code meets all the requirements and is functioning as expected.*

■ Q3. Write a Java program that reads an integer from stdin and checks if it is a Fibonacci number. **Input Format:** - Line 1: An integer **Output Format...

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int a = 0, b = 1;

        while (b < n) {
            int temp = a + b;
            a = b;
            b = temp;
```

```
        }

        if (n == a || n == b) {
            System.out.println("Fibonacci");
        } else {
            System.out.println("Not Fibonacci");
        }
    }
}
```

■ *Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and effective implementation. Great job on achieving an "Accepted" status, indicating that your solution meets all requirements.*

■ Q4. Write a Java program that reads a string from stdin and validates it using an enum. **Input Format:** - Line 1: A string **Output Format:** - "Valid...

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        solution();
    }

    public static void solution() {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();

        if(input.equals("hello")) {
            System.out.println("Valid");
        } else {
            System.out.println("Invalid");
        }
    }
}
```

■ *Your code has successfully passed all test cases, indicating a correct and effective implementation of the solution. Great job on achieving an "Accepted" status, keep up the good work!*

■ Q5. Write a Java program that reads a string from stdin and encrypts it using a simple substitution cipher. **Input Format:** - Line 1: A string **Outpu...

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();

        StringBuilder result = new StringBuilder();

        for (char c : input.toCharArray()) {
            char encrypted = (char)(c + 3);
            result.append(encrypted);
        }
```

```
        System.out.println(result.toString());
    }
}
```

■ *Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on achieving an "Accepted" status, keep up the good work!*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.